MIL-M-50079D
30 SEPTEMBER 1964
SUPERSEDING
MIL-M-50079C(MU)
4 JUNE 1964

MILITARY SPECIFICATION

MASK, PROTECTIVE, FIELD, ABC-M17

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of protective mask with filter elements in the facepiece.

1.2 Classification. Masks shall be of the following sizes, as specified (see 6.2):

Small
Medium
Large

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-601 — Boxes, Wood, Cleated-Plywood.

PPP-B-636 — Box, Fiberboard.

PPP-T-76 — Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

MILITARY

MIL-P-116 — Preservation, Methods of.

MIL-B-131 — Barrier Material; Water Vaporproof, Flexible.

MIL-B-13239 — Barrier Material, Waterproofed, Flexible, All Temperature.

MIL-F-50078 — Filter Element, Protective Mask, M13.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-282 — Filter Units, Protective Clothing, Gas Mask Components, and Related Products; Performance Test Methods.

FSC 4240

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DRAWINGS

EDGEWOOD ARSENAL

DL5-1-293 — Mask, Protective, Field, ABC-M17.

(All drawings referenced in this specification are listed on Drawing DL 5-1-293.)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Union Chicago Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 **Materials and components.** All materials and components shall conform to the specifications and drawings forming a part of this specification.

3.2 **Assembly.** Assembly shall be in accordance with Drawing E5-1-293.

3.2.1 *Filter elements.* Only matched pairs of filter elements shall be assembled into finished masks (see MIL-F-50073).

3.3 **Leakage.**

3.3.1 *Mask assembly leakage.* Leakage of the mask assembly (excluding leakage at the periphery, outlet-valve seat, and through filter elements) shall not exceed 0.003 percent DOP smoke penetration when tested as specified in 4.6.1.

3.3.2 *Outlet-valve leakage.* Leakage of the outlet valve shall not exceed 15 milliliters of air per minute at a pressure differential of 25 milliliters of water when tested as specified in 4.6.2.

3.3.3 *Package leakage and heat seal.* Package leakage and heat seal tests shall be conducted on packaged mask assemblies as specified in 4.6.7.

3.4 **Resistance.**

3.4.1 *Inhalation resistance.* Inhalation resistance of the mask assembly shall not exceed 55 millimeters of water at a continuous airflow of 85 liters per minute when tested as specified in 4.6.3.

3.4.2 *Exhalation resistance.* The exhalation resistance of the mask assembly shall not exceed 23 millimeters of water at a continuous airflow of 85 liters per minute when tested as specified in 4.6.4.

3.5 **Filter elements.**

3.5.1 *Moisture content of filter elements in mask.* The average moisture content of the filter elements in the packaged mask shall not exceed 2.0 percent and no individual element shall exceed 2.5 percent moisture when tested as specified in 4.6.5.

3.5.2 *DOP smoke penetration.* After moisture content testing is completed, the same filter elements shall be tested for DOP smoke penetration. The DOP smoke penetration of each filter element shall not exceed 0.015 percent when tested as specified in 4.6.6.

3.6 **Workmanship.** The mask assembly shall be free from foreign matter, and damage such as permanent set of rubber parts, cracks, dents, tears, and malfunctioning of hardware.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to specified requirements.

4.2 Special provisions.

4.2.1 *Alternative inspection.* The supplier may utilize any alternative inspection procedure which will provide equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Objective evidence.* The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2.3 *Inspection substation.* If the supplier elects to establish inspection substations for the purpose of conducting inspection for certain listed defects, the Acceptable Quality Levels (AQL's) for each inspection substation shall be equal to the AQL which is the next numerically lower value than the AQL specified for the total listed defects. The requirements of MIL-STD-105 are applicable to each substation and to the end item.

4.3 Lotting. A regular production lot shall consist of the masks of one size produced by one manufacturer, at one plant from the same materials and under essentially the same manufacturing conditions. However, no more than one lot of filter elements, faceblanks, voicemitter outlet-valve assemblies, or nosecups shall be represented in any one lot of finished masks.

4.4 Sampling.

4.4.1 *For examination and nondestructive tests.* Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2 *For destructive tests.* Sampling for 4.6.5, 4.6.6, and 4.6.7 will be conducted in accordance with MIL-STD-105, using level S-3.

4.5 Inspection provisions.

4.5.1 *For examination and nondestructive tests.* Sample masks from the regular production lots shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.5.2 *For destructive tests.* Samples taken in accordance with 4.4.2 shall be tested as specified in 4.6.5, 4.6.6, and 4.6.7. The acceptance number is 0.

4.5.3 *For critical defects.* Each item in the lot shall be inspected for critical characteristics listed in the classification of defects.

4.5.4 *Classification of defects.*

4.5.4.1 *Mask assembly (Dwg E5-1-293).*

Categories	Defects	Acceptance standards
Critical:		
1	Mask assembly leakage	4.6.1
2	Outlet valve leakage	4.6.2
3	Lens surface split or cracked.	

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Categories	Defects	Acceptance standards
Major:	AQL 1.0 percent defective.	
101	Component missing, incorrect, or incorrectly assembled.	
102	Workmanship (3.6).	
103	Inlet valve assembly inoperable.	
104	Marking incorrect.	
105	Unmatched pair of filter elements.	
196	Head harness buckle not free to rotate in loop of clip.	
Major A:	AQL 0.25 percent defective.	
150	Inhalation resistance	4.6.3
151	Exhalation resistance	4.6.4
Minor:	AQL 2.5 percent defective.	
201	Protective finish missing or incorrect.	
202	Marking missing or illegible.	

4.6 Tests.

4.6.1 *Mask assembly leakage.* Leakage of the mask assembly, complete with filter elements and inlet valve assembly, shall be determined using Edgewood Arsenal Q14 Gas Mask Leakage Tester. Each eyepiece assembly and voicemitter assembly shall be flexed by hand during the testing operation.

4.6.2 *Outlet valve leakage.* The leakage of the outlet valve shall be determined using Edgewood Arsenal M4A1 Outlet Valve Leakage Indicator.

4.6.3 *Inhalation resistance.* Inhalation resistance of the mask assembly shall be determined using Edgewood Arsenal Q43 Inhalation and Exhalation Resistance Indicator.

4.6.4 *Exhalation resistance.* Exhalation resistance of the mask assembly shall be determined using Edgewood Arsenal Q43 Inhalation and Exhalation Resistance Indicator.

4.6.5 *Moisture content of filter elements.* Weigh the filter elements from each sample mask to the nearest 0.1 gram (net weight) immediately after unpackaging the mask. Place the elements in a forced-air drying oven at 160° ± 5°F. and a maximum of 5 percent relative humidity until constant dry weight, within 0.1 gram, is reached. Record the final weighing (dry weight) to the nearest 0.1 gram and determine the percent moisture content by the following formula:

$$\text{Percent moisture content} = \frac{100\ (\text{wet weight} - \text{dry weight})}{\text{wet weight})}$$

4.6.6 *DOP smoke penetration.* DOP smoke penetration of the filter elements shall be determined in accordance with method 502.1.1 of MIL-STD-282 at a flow rate of 16 liters per minute.

4.6.7 *Package leakage and heat seal test.* Leakage and heat seal characteristics of the sealed water-vaporproof bag containing mask assembly shall be determined by subjecting the samples taken in accordance with 4.4.2 to the quick leak and heat seal test of MIL-P-116.

5. PREPARATION FOR DELIVERY (For repair parts packaging see 6.4)

5.1 Packaging, level A.

5.1.1 *Type I mask packaging (see 6.1.2 and 6.2).* No preservative compounds shall be applied to any portion of the mask or its

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components. Each mask, with a faceform positioned in the mask as shown on Drawing E5-1-293, shall be placed in a water-vaporproof heat-sealable bag fabricated from barrier-material conforming to Class 1 of MIL-B-131, except that the heat-sealable surface of the material shall be a nonplasticized type, such as polyethylene. The package shall be in conformance with the requirements of method IA-8 of MIL-P-116, except that sampling shall be conducted in accordance with 4.4.2 of the mask specification. Water-vaporproof bags cited above shall have the same inspection acceptance criteria as the packaged mask (see 4.4.2). Cushioning, shall be nonhygroscopic, nondusting and compatible with the mask. Each packaged mask shall be placed in a canvas carrier as shown on Drawing E5-1-293. The canvas carrier containing the packaged mask shall then be placed in a W5c or W6c, style RSC, fiberboard box, conforming to PPP-B-636. In addition to closure requirements for slotted style interior boxes the box shall be taped as specified, except that pressure-sensitive tape conforming to PPP-T-76 shall be used.

5.1.2 *Type II mask packaging (see 6.1.2 and 6.2).* The mask shall be packaged in the same manner as specified in 5.1.1, except that the fiberboard box shall be omitted. In lieu thereof, each canvas carrier containing a packaged mask shall be placed in a waterproof heat-sealable bag fabricated from barrier material, conforming to type B2 (stretchable or nonstretchable), class 1, grade A or B; or type B-3, class 1, grade A or B of MIL-B-13239. The package shall be in conformance with the requirements of method IC-3 of MIL-P-116, except that the package leakage and heat seal test need not be performed.

5.2 Packing, level A.

5.2.1 *Type I, mask packing (see 6.1.2 and 6.2).* Ten boxes of masks, of one size, packaged as specified in 5.1.1, shall be packed in a V3c, style RSC, fiberboard box conforming to PPP-B-636. In addition to closure requirements, for slotted exterior boxes, the box shall be taped as specified. Tape banding shall be used in lieu of metal banding to reinforce box.

5.2.2 *Type II, mask packing (see 6.1.2 and 6.2).* Ten boxes of masks, of one size, packaged as specified in 5.1.2 shall be packed in a snug-fitting V3s, style FTC fiberboard box, conforming to PPP-B-636, without box cover. The box shall be divided into 10 cells of the same size extending the full depth of the box by means of interlocking slotted partitions (see fig. 1) made from the same material as the box. The extended ends of the partitions shall be stapled or stitched to the sides, ends and bottom of the box with staples or wire in accordance with the requirements of PPP-B-636 for the securing of the manufacturer's joint. The interlocking partitioned fiberboard box with the packaged masks packed therein, shall be placed in a snug-fitting, oversea-type, style A, wood, cleated-plywood box, using type II, class 2, plywood in conformance with PPP-B-601. The cover of the cleated plywood box shall not be nailed in place, but shall be secured to the box with cleated-box fasteners covered by Federal Stock No. 5540-600-6223, using two fasteners on each side and one fastener at each end of the box. The fasteners shall be assembled to the box in accordance with the instructions of the fastener manufacturer. In addition, the cleated plywood box shall be strapped with flat galvanized metal strapping in conformance with the appendix to PPP-B-601. The cleated-box fasteners on the sides of the box shall be placed over the strapping.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 *Marking for type II pack.* The following additional markings shall be stenciled in

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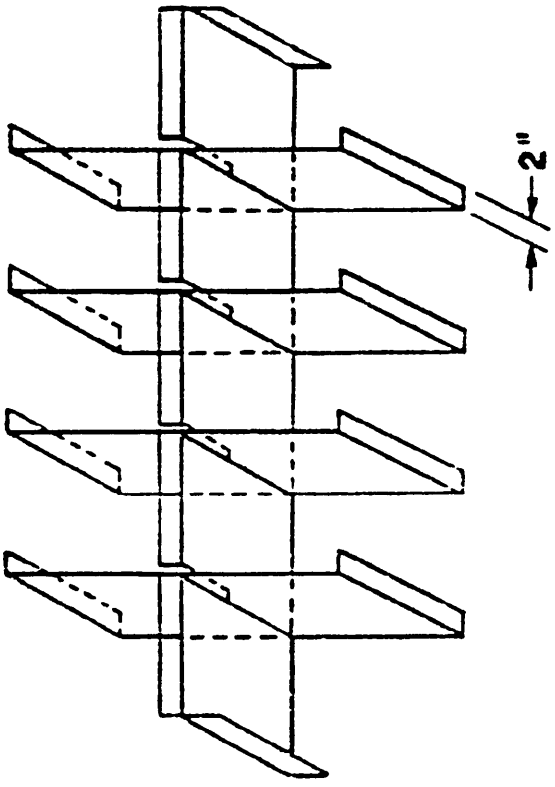


Figure 1 - Interlocking slotted partition

minimum ¾-inch high letters on the underside of the box cover.

BOX IS FOR USE IN STORING MASKS
BY GARRISON OR FIELD PERSONNEL

6. NOTES

6.1 Intended use.

6.1.1 *Mask.* The mask covered by this specification is intended for protection of the eyes and respiratory system against toxicological warfare agents. It is not intended for protection against carbon monoxide or ammonia.

6.1.2 *Packaging and packing (see section 5).*

6.1.2.1 *Type I, package and pack.* Type I is intended as a nonreusable package and pack for use in the shipment and storage of protective masks.

6.1.2.2 *Type II, package and pack.* Type II packaging and packing has been designed for use (and garrison or field reuse, when required) as a dual-purpose shipping container and as a device for the protection and orderly garrison (or field) storage of masks during periods when masks which have been issued to troops are not carried by them (see TM3-4240-202-25P and TM3-4240-202-15, Mask, Protective, Field, ABC-M17).

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Size of mask.

(c) Elements for lot numbering (see Directive 511).

(d) Select type I packaging and packing or type II packaging and packing.

6.3 **Component specifications.** Component specifications for the M17 mask are:

Title	*Number*
Backing Material, C18	MIL-B-51097
Cap-Inlet Valve Assembly, C20	MIL-C-50081
Carrier, Field Protective Mask, M15	MIL-C-50062
Core Material, C18	MIL-C-51096
Cover, Voicemitter-Outlet Valve, C13	MIL-C-50072
Discs, Inlet and Nosecup Valve	MIL-D-50082
Discs, Outlet Valve	MIL-D-10132
Eyering, Field Protective Mask, C4	MIL-E-50069
Faceblank, C17	MIL-F-50070

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Title	Number
Faceform, Field Protective Mask, M8	MIL-F-50073
Filter Element, Protective Mask, M13	MIL-F-50078
Filter Material, Gas-Aerosol, C18	MIL-F-51095
Head harness, C8R1	MIL-H-10120
Lenses, Ophthalmic, Simple (For ABC-M17 Mask)	MIL-L-50064
Nosecup, C10	MIL-N-50071
Outsert, Eyelens, M1	MIL-O-50077
Ring, Crimping (For ABC-M17 Mask)	MIL-R-50068
Screen, Flocked Grid	MIL-S-50080
Valve, Nosecup	MIL-V-50065
Voicemitter-Outlet Valve, Assembly, C29R1	MIL-V-50066
Voicemitter-Outlet Valve, C29R1; Components for	MIL-V-50067

6.4 **Repair parts packaging.** When repair parts packaging is required for items authorized for stocking as repair parts, the repair parts packaging data sheets covering the applicable components of the M17 Mask should be used. (Copies of repair parts packaging data sheets may be obtained from the Director for Engineering and Industrial Services, ATTN: Chief, Design Branch, U.S. Army Edgewood Arsenal, Edgewood Arsenal, Md. 21010.)

6.5 **International interest.** Certain provisions of this specification are the subject of ABC international standardization agreement. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

Custodians:

Army—MU

Navy—SH

Air Force—84

International interest (see 6.5).

Review activities:

Army—MD

Navy—SH

Air Force—84

User activity:

Navy—MC

Preparing activity:

Army—MU(EA)

Project No. 4240-0068

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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